

Capturing the nuances of the grammaticalisation of perfects

2. An under-considered nuance in the grammaticalisation of perfects. Morphological perfects whose diachronic source was a copula *have/be* + past participle (pp) construction are found in multiple Indo-European languages. The broader contours of their grammaticalisation are well-documented (e.g. Bybee et al. 1994): (i) a state-denoting RESULTATIVE marker develops into (ii) a temporally present, aspectual ANTERIOR marker, before developing into (iii) a temporal PAST marker. Indeed, closely related languages are at different stages in this grammaticalisation, e.g. English, unlike German and Dutch, lacks (iii). Pursuing an upwards reanalysis approach (e.g. Roberts & Roussou 1999) based on Cinque's universal hierarchy, IJbema (2002) proposes that (1) captures the changes in (i) to (iii).

Supporting evidence for Dutch is in (2): As auxiliary *have* can both embed, (2a), and be embedded by, (2b), the modal *moet(en)* ‘must’ (assumed to lexicalise Mod_{obligation}), it can occupy two structural positions - T(Past) and T(Anterior)/Asp_{perfect} respectively (IJbema 2002: 56-60). In turn, the possibility of (2b) but not (2a) in English correctly predicts the availability of (ii) only, whilst Dutch and German *have* both (2a/b) and allow both (ii/iii) (Wall 2018).

However, an important issue which (1) does not address concerns the different types of ANTERIORS, as documented by the rich cross-linguistic literature (e.g. Comrie 1976). In particular, languages like Dutch, German and English distinguish both perfects-of-result (PORs; $\approx$ present state resulting from an instantiated event) and experiential perfects (EPs; $\approx$ potential plurality of events in time span up to present). That the PoR-EP distinction is syntactically relevant is supported by languages which employ morphologically distinct auxiliaries for EPs/PoRs such as Icelandic (e.g. McFadden 2007), and those which have the PoR but lack the EP like Hiberno-English (Harris 1984). Crucially, it has also been proposed (e.g. Slobin 1994; Croft 2012: 143), that EPs represent an advanced grammaticalisation step compared with PoRs. However, if there is only one structural position available for all ANTERIORS, the change from PoR to EP clearly defies an upward reanalysis explanation.

3. PoRs & RESULTATIVES and EPs & PASTS as semantic counterparts. Whilst I will maintain that PoRs and EPs are structural counterparts with each other but not with RESULTATIVES/PASTS, I propose here that PoRs semantically parallel RESULTATIVES, and EPs semantically parallel PASTS. More specifically, PoRs/RESULTATIVES exhibit parallel modification restrictions which are lacking in EPs/PASTS. Mittwoch (2008: 328-330) shows that manner adverbs, instrumentals, and spatial modifiers, e.g. (3a), are all ruled out in active PoRs unless relating to the state. A novel observation is that these correspond to well-documented restrictions on the same elements in RESULTATIVES like (3b) (e.g. Rapp 1998).

- (3) a. *You've corrected the proofs #too slowly / with red ink.* (based on Mittwoch 2008: 329)
 b. *Der Brief ist *langsam/ mit roter Tinte geschrieben.* (Rapp 1998: 257)
 The letter is slowly with red ink written.PTCP

In contrast, such modification is allowed in EPs, which also allow quantificational adverbs like *mehrmals* 'multiple times', unlike PoRs (Mittwoch 2008). I further show that PASTS, disambiguated from PoRs/EPs by the acceptability of deictic past tense adverbs like *gestern* 'yesterday', parallel EPs in both respects, as shown in (4) (ex. based on Rapp 1998: 246).

- (4) *Gestern hat Tobias den Mülleimer mehrmals / mit der Heugabel geleert.*
 Yesterday has Tobias the bin multiple.times with the pitchfork emptied.PTCP

4. Different markers have distinct heads and distinct semantic units. I propose that these semantic parallels are explained by the distinct lexical semantics of the auxiliaries/copulas: those in PoRs and RESULTATIVES are [+perf(ective)] ($\approx$ non-inherent state/characteristic) whilst those in EPs and PASTS are [-perf(ective)] ($\approx$ inherent state/characteristic). These respective semantics are independently attested in the lexical guises of *be/have* like those in (5).

- (5) a. *John is happy (today).* b. *John has a book.* c. *John is tall.* d. *John has blue eyes.*
 Being non-inherent, [+perf] instances of *be/have* like (5a/b) impose a specific starting point on the state/characteristic expressed, whilst being inherent, the [-perf] instances in (5c/d) lack this. I propose that the restrictions on non-state relating modifiers in PoRs/RESULTATIVES follow from this difference.

To represent the differences between markers formally, I assume, based on Wiltschko (2014), that these markers are categories (*c*) which are pairings of a phonological unit (π) and a semantic unit (*E*). Abstracting away from Wiltschko's universal categories, I assume these are in turn paired with a particular structural head, (*h*), i.e. $c = \langle h, \{\pi, E\} \rangle$. The resultant categories corresponding to the lexical guises of *be/have* in (5) are shown in (6):

- (6) a. $C:\text{LEXICAL}[+\text{PERF}] = \langle V, \{\pi:be/have, E:[+\text{perf}]\} \rangle$ b. $C:\text{LEXICAL}[-\text{PERF}] = \langle V, \{\pi:be/have, E:[-\text{perf}]\} \rangle$

Largely parallel to IJbema's (1), I further propose the following categories for RESULTATIVES, PoRs, EPs, and PASTS:

- (7) a. $C:\text{RESULTATIVE} = \langle \text{Voice}, \{\pi:be/have, E:[+\text{perf}]\} \rangle$ b. $C:\text{PoR} = \langle \text{Aspect}, \{\pi:be/have, E:[+\text{perf}]\} \rangle$
 c. $C:\text{EP} = \langle \text{Aspect}, \{\pi:be/have, E:[-\text{perf}]\} \rangle$ d. $C:\text{PAST} = \langle \text{Tense}, \{\pi:be/have, E:[-\text{perf}]\} \rangle$

5. Semantic change parallels upwards reanalysis and phonological change. Based on the proposed categories, the change from PoR to EP is now explained by a reanalysis of the relevant semantic unit, i.e. the substitution of *c:PoR's E* from [+perf] to [-perf] (i.e. (7b) to (7c)). Two core properties of this semantic reanalysis are as follows. Firstly, it directly parallels both phonologically/morphologically-induced reanalysis (i.e. substitution of π) and upwards grammaticalisation (i.e. substitution of *h*), and is an equally minimal process. Secondly, it does not involve the creation of new meaning but rather the re-assignment of the [-perf] meaning present in the corresponding lexical category, (6b). I propose that what actually triggers this semantic reanalysis are **semantic changes** in the lexical categories. More specifically, I hypothesise that it is enabled by a growth in the frequency of (6b). With regards to the Regress Problem this has the following strength: In contrast to the invariance of grammar, semantic change is ubiquitous - new meanings are easy to gain and acquired throughout a speaker's lifetime (see e.g. Rodd 2018). In other words, post-acquisition state changes in C_1 are commonplace when we are dealing with semantic units.

References: • Bybee, J., Perkins, R., & W. Pagliuca. 1994. *The evolution of grammar: Tense, aspect, and modality in the languages of the world*. Chicago: University of Chicago Press. • Comrie, B. 1976. *Aspect*. Cambridge: Cambridge University Press. • Croft, W. 2012. *Verbs: Aspect and causal structure*. Oxford: Oxford University Press. • Harris, J. 1984. "Syntactic variation and dialect divergence." *Journal of linguistics*, 20(2), 303-327. • IJbema, A. 2002. *Grammaticalization and infinitival complements in Dutch*. Utrecht: LOT Publications. • Kroch, A. 2000. "Syntactic Change". In: Baltin, M. & C. Collins (eds.), *The handbook of contemporary syntactic theory*. Oxford: Blackwell. • Lundquist, B. 2008. *Nominalizations and participles in Swedish*. Ph.D. thesis, University of Tromsø. • McFadden, T. 2007. "Auxiliary selection." *Language and linguistics compass* 1(6), 674-708. • Mittwoch, A. 2008. "The English resultative perfect and its relationship to the experiential perfect and the simple past tense." *Linguistics and Philosophy* 31(3), 323-251. • Rapp, I. 1998. "Zustand? Passiv?—Überlegungen zum sogenannten „Zustandspassiv“." *Zeitschrift für Sprachwissenschaft* 15(2), 231-265. • Roberts, I., & A. Roussou. 1999. "A formal approach to "grammaticalization"." *Linguistics*, 37(6), 1011-1041. • Roberts, I. 2007. *Diachronic syntax*. Oxford: Oxford University Press. • Rodd, J. 2018. "Lexical ambiguity." In: S.-A. Rueschemeyer & M. G. Gaskell (eds.) *The Oxford handbook of psycholinguistics* (2 ed.) Oxford: Oxford University Press. • Slobin, D. 1994. "Talking perfectly." In: *Discourse origins of the present perfect*. Amsterdam: John Benjamins. • Wall, J. 2018. *Seeing double: The HAVE puzzle*. MA thesis, Utrecht University. • Wiltschko, M. 2014. *The universal structure of categories: Towards a formal typology*. Cambridge: Cambridge University Press. •